

MongoDb: Atelier 6

Exercice 1 — Se connecter à MongoDB en Python :

```
ateliers > public > atelier6.py
1  from pymongo import MongoClient
2  client = MongoClient("mongodb://localhost:27017/")
3  db = client["atelierPythonMongo"]
4  print("Base connectée !")
5
6  |
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\OneDrive\Desktop\Atelier> & C:/Python314/python.exe c:/Users/dell/OneDrive/Desktop/Atelier/ateliers/public/atelier6.py
Base connectée !
PS C:\Users\dell\OneDrive\Desktop\Atelier>
```

Exercice 2 — Se connecter à MongoDB en Python :

on utilise `print(db.list_collection_names())`

```
6  print(db.list_collection_names())
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dell\OneDrive\Desktop\Atelier> & C:/Python314/python.exe c:/Users/dell/OneDrive/Desktop/Atelier/ateliers/public/atelier6.py
Base connectée !
['utilisateurs', 'commandes', 'produits']
```

Exercice 3 — Lire les utilisateurs :

for u in `db.utilisateurs.find(): print(u)`

```
['utilisateurs', 'commandes', 'produits']
{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 23, 'ville': 'Tanger', 'hobbies': ['football', 'gaming']}
{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Sara', 'email': 'sara@test.com', 'age': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}
{'_id': ObjectId('691d844796939f3a72697bf4'), 'nom': 'Nadia', 'email': 'nadia@test.com', 'age': 35, 'ville': 'Casa', 'hobbies': ['cuisine']}
{'_id': ObjectId('691d844796939f3a72697bf5'), 'nom': 'Yassine', 'email': 'yassine@test.com', 'age': 27, 'ville': 'Fès', 'hobbies': ['fitness', 'musique']}
{'_id': ObjectId('691d844796939f3a72697bf6'), 'nom': 'Hassan', 'email': 'hassan@test.com', 'age': 31, 'ville': 'Tanger', 'hobbies': ['lecture']}
PS C:\Users\dell\OneDrive\Desktop\Atelier>
```

Exercice 4 — Insérer un nouvel utilisateur via Python :

on utilise `db.utilisateurs.insert_one({ })`

```
9 db.utilisateurs.insert_one({
10     "nom": "Karim",
11     "email": "karim@test.com",
12     "age": 26,
13     "ville": "Agadir",
14     "hobbies": ["surf", "musique"]
15 })
16
17
18
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
print(u)
['utilisateurs', 'commandes', 'produits']
{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 23, 'ville': 'Tanger', 'hobbies': ['football', 'gaming']}
{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Sara', 'email': 'sara@test.com', 'age': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}
{'_id': ObjectId('691d844796939f3a72697bf4'), 'nom': 'Nadia', 'email': 'nadia@test.com', 'age': 35, 'ville': 'Casa', 'hobbies': ['cuisine']}
{'_id': ObjectId('691d844796939f3a72697bf5'), 'nom': 'Yassine', 'email': 'yassine@test.com', 'age': 27, 'ville': 'Fès', 'hobbies': ['fitness', 'musique']}
{'_id': ObjectId('691d844796939f3a72697bf6'), 'nom': 'Hassan', 'email': 'hassan@test.com', 'age': 31, 'ville': 'Tanger', 'hobbies': ['lecture']}
PS C:\Users\dell\OneDrive\Desktop\Atelier>
```

Exercice 5 — Filtrage (find) :

`for u in db.utilisateurs.find({"age": {"$gt": 28}}): print(u)`

```

17 for u in db.utilisateurs.find({"age": {"$gt": 28}}):
18     print(u)
19
20
21

```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```

PS C:\Users\dell\OneDrive\Desktop\Atelier> & C:/Python314/python.exe c:/Users/dell/OneDrive/Desktop/Atelier/ateliers/public/atelier6.py
{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Sara', 'email': 'sara@test.com', 'age': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}
{'_id': ObjectId('691d844796939f3a72697bf4'), 'nom': 'Nadia', 'email': 'nadia@test.com', 'age': 35, 'ville': 'Casa', 'hobbies': ['cuisine']}
{'_id': ObjectId('691d844796939f3a72697bf6'), 'nom': 'Hassan', 'email': 'hassan@test.com', 'age': 31, 'ville': 'Tanger', 'hobbies': ['lecture']}

```

Exercice 6 — Mise à jour :

Modifier l'âge d'Ali : `db.utilisateurs.update_one({"nom": "Ali"}, {"$set": {"age": 24}})`

```

19 db.utilisateurs.update_one(
20     {"nom": "Ali"},
21     {"$set": {"age": 24}}
22 )
23
24
25

```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```

r/ateliers/public/atelier6.py
['utilisateurs', 'commandes', 'produits']
{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 24, 'ville': 'Tanger', 'hobbies': ['football', 'gaming']}

```

Ajouter un hobby : `db.utilisateurs.update_one({"nom": "Ali"}, {"$push": {"hobbies": "cinéma"}})`

```

['utilisateurs', 'commandes', 'produits']
{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 24, 'ville': 'Tanger', 'hobbies': ['football', 'gaming', 'cinéma']}

```

Exercice 7 — Supprimer :

on utilise `db.utilisateurs.delete_one({"nom": "Hassan"})`

```

29 db.utilisateurs.delete_one({"nom": "Hassan"}) |
30
31
32

```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

+ v

```

['utilisateurs', 'commandes', 'produits']
{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 24, 'ville': 'Tanger', 'hobbies': ['football', 'gaming', 'cinéma', 'cinéma', 'cinéma']}
{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Sara', 'email': 'sara@test.com', 'age': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}
{'_id': ObjectId('691d844796939f3a72697bf4'), 'nom': 'Nadia', 'email': 'nadia@test.com', 'age': 35, 'ville': 'Casa', 'hobbies': ['cuisine']}
{'_id': ObjectId('691d844796939f3a72697bf5'), 'nom': 'Yassine', 'email': 'yassine@test.com', 'age': 27, 'ville': 'Fès', 'hobbies': ['fitness', 'musique']}
{'_id': ObjectId('691d8da03026f8f402a1768c'), 'nom': 'Karim', 'email': 'karim@test.com', 'age': 26, 'ville': 'Agadir', 'hobbies': ['surf', 'musique']}

```

Exercice 8 — Index :

on utilise `db.utilisateurs.create_index([("email", 1)], unique=True)`

```

db.utilisateurs.create_index([("email", 1)], unique=True)
print("tres bien")

```

Exercice 9 — Agrégation (statistiques) :

```

35 pipeline = [
36     {"$group": {"_id": "$ville", "age_moyen": {"$avg": "$age"}}},
37     {"$sort": {"age_moyen": -1}}
38 ]
39 for doc in db.utilisateurs.aggregate(pipeline):
40     print(doc)
41
42

```

```

{'_id': 'Casa', 'age_moyen': 35.0}
{'_id': 'Rabat', 'age_moyen': 29.0}
{'_id': 'Fès', 'age_moyen': 27.0}
{'_id': 'Agadir', 'age_moyen': 26.0}
{'_id': 'Tanger', 'age_moyen': 24.0}
PS C:\Users\de11\OneDrive\Desktop\Atelier>

```

Exercice 10 — Jointure : afficher commandes + utilisateurs + produits :

```
42 pipeline = [
43     {
44         "$lookup": {
45             "from": "utilisateurs",
46             "localField": "idUtilisateur",
47             "foreignField": "_id",
48             "as": "detailsUtilisateur"
49         }
50     },
51     {
52         "$lookup": {
53             "from": "produits",
54             "localField": "idProduit",
55             "foreignField": "_id",
56             "as": "detailsProduit"
57         }
58     }
59 ]
60 for doc in db.commandes.aggregate(pipeline):
61     print(doc)
62
63
```

```
{ '_id': ObjectId('691d84a596939f3a72697bf5'), 'idUtilisateur': ObjectId('691d844796939f3a72697bf2'), 'idProduit': ObjectId('691d845796939f3a72697bf7'), 'quantite': 1, 'dateCommande': datetime.datetime(2024, 2, 1, 0, 0), 'detailsUtilisateur': [{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 24, 'ville': 'Tanger', 'hobbies': ['football', 'gaming', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma']}, {'_id': ObjectId('691d845796939f3a72697bf7'), 'nom': 'PC Portable', 'prix': 7000, 'stock': 12}]}{'_id': ObjectId('691d84a596939f3a72697bfc'), 'idUtilisateur': ObjectId('691d844796939f3a72697bf2'), 'idProduit': ObjectId('691d845796939f3a72697bf8'), 'quantite': 2, 'dateCommande': datetime.datetime(2024, 2, 10, 0, 0), 'detailsUtilisateur': [{'_id': ObjectId('691d844796939f3a72697bf2'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 24, 'ville': 'Tanger', 'hobbies': ['football', 'gaming', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma']}, {'_id': ObjectId('691d845796939f3a72697bf8'), 'nom': 'Souris Gaming', 'prix': 250, 'stock': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}], 'detailsProduit': [{'_id': ObjectId('691d845796939f3a72697bf8'), 'nom': 'Souris Gaming', 'prix': 250, 'stock': 50}]}
```

```
{ '_id': ObjectId('691d84a596939f3a72697bfe'), 'idUtilisateur': ObjectId('691d844796939f3a72697bf3'), 'idProduit': ObjectId('691d845796939f3e'), 'quantite': 29, 'dateCommande': datetime.datetime(2024, 2, 29, 0, 0), 'detailsUtilisateur': [{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Ali', 'email': 'ali@test.com', 'age': 24, 'ville': 'Tanger', 'hobbies': ['football', 'gaming', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma', 'cinéma']}, {'_id': ObjectId('691d845796939f3e'), 'nom': 'Souris Gaming', 'prix': 250, 'stock': 50}]}
```

```
{ '_id': ObjectId('691d84a596939f3a72697bfe'), 'idUtilisateur': ObjectId('691d844796939f3a72697bf3'), 'idProduit': ObjectId('691d845796939f3a72697bf7'), 'quantite': 1, 'dateCommande': datetime.datetime(2024, 3, 12, 0, 0), 'detailsUtilisateur': [{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Sara', 'email': 'sara@test.com', 'age': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}], 'detailsProduit': [{'_id': ObjectId('691d845796939f3a72697bf7'), 'nom': 'PC Portable', 'prix': 7000, 'stock': 12}]}
```

```
{ '_id': ObjectId('691d84a596939f3a72697bff'), 'idUtilisateur': ObjectId('691d844796939f3a72697bf3'), 'idProduit': ObjectId('691d845796939f3a72697bf8'), 'quantite': 3, 'dateCommande': datetime.datetime(2024, 3, 20, 0, 0), 'detailsUtilisateur': [{'_id': ObjectId('691d844796939f3a72697bf3'), 'nom': 'Sara', 'email': 'sara@test.com', 'age': 29, 'ville': 'Rabat', 'hobbies': ['lecture', 'voyage']}], 'detailsProduit': [{'_id': ObjectId('691d845796939f3a72697bf8'), 'nom': 'Souris Gaming', 'prix': 250, 'stock': 50}]}
```

```
PS C:\Users\dell\OneDrive\Desktop\Atelier>
```